

EVELETH-VIRGINIA, MN, USA

WMO: 727474

Lat: 47.424N

Lon: 92.498W

Elev: 1379

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-17.1	-29.4	1.1	-22.0	-24.2	1.5	-15.4	20.6	13.7	18.7	12.7	2.4	310	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	85.5	69.1	81.9	66.7	79.9	65.3	72.7	81.2	70.4	78.5	68.3	76.4	10.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	116.7	77.8	67.8	107.4	75.6	64.8	96.7	72.5	37.2	81.0	35.2	78.0	33.4	75.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	17.4	15.5	DB	-30.4	90.2	7.2	3.4	-35.6	92.7	-39.8	94.7	-43.8	96.6	-49.0	99.0	(4)
(5)			WB	-30.4	76.1	6.4	2.3	-35.0	77.8	-38.8	79.2	-42.4	80.5	-47.1	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.4	8.9	13.2	25.8	39.1	51.5	61.0	65.9	63.6	55.7	43.1	28.5	15.2		
	DBStd	22.12	14.17	13.13	12.70	9.12	8.00	6.80	5.74	5.77	7.97	8.15	10.62	12.62		
	HDD50	5475	1273	1030	751	340	75	4	0	0	32	247	644	1079		
	HDD65	9526	1738	1450	1214	777	427	155	58	96	293	680	1094	1544		
	CDD50	1624	0	0	2	14	121	334	494	421	204	33	1	0		
	CDD65	199	0	0	0	0	8	35	87	52	16	1	0	0		
	CDH74	2518	0	0	0	9	163	482	1036	626	188	14	0	0		
	CDH80	529	0	0	0	0	29	104	248	119	28	1	0	0		
	WSAvg	6.2	6.1	6.3	6.5	6.8	6.7	5.7	5.4	5.3	5.8	6.7	6.7	6.0		
	PrecAvg	27.1	0.9	0.6	1.1	1.9	2.9	4.3	4.1	3.5	3.3	2.6	1.3	0.8		
(15) Precipitation	PrecMax	34.4	3.3	1.9	3.0	5.4	5.9	8.4	11.9	8.7	6.7	7.0	3.7	2.0		
	PrecMin	16.1	0.2	0.0	0.1	0.3	0.4	1.2	1.1	0.6	0.7	0.3	0.1	0.1		
	PrecStd	3.7	0.7	0.4	0.6	1.2	1.3	1.7	2.4	2.0	1.5	1.7	0.9	0.5		
	PrecStd	3.7	0.7	0.4	0.6	1.2	1.3	1.7	2.4	2.0	1.5	1.7	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	45.7	63.2	73.1	83.8	88.3	90.3	87.9	83.5	75.3	60.4	41.5		
		MCWB	36.0	39.4	51.4	54.7	64.0	72.1	72.2	71.2	67.1	59.7	50.6	35.8		
	2%	DB	35.5	40.6	54.3	66.0	79.1	82.2	85.8	82.3	79.4	67.9	54.0	37.3		
		MCWB	33.2	36.1	45.2	49.9	60.1	66.6	70.1	67.3	65.2	56.1	47.6	34.6		
	5%	DB	32.1	36.3	47.6	62.4	73.2	79.3	82.1	80.8	74.7	62.8	46.4	34.4		
		MCWB	31.0	33.2	41.1	47.1	56.9	64.1	67.5	66.6	63.1	53.9	41.5	32.9		
	10%	DB	28.3	32.4	43.0	57.2	68.3	76.5	80.1	78.6	72.0	58.5	43.1	32.3		
		MCWB	26.7	30.3	37.1	44.3	54.6	62.8	66.2	65.2	61.9	50.6	38.7	31.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.7	40.2	55.4	57.5	67.7	74.7	75.7	74.7	71.5	62.9	53.7	38.2		
		MCDB	40.4	44.2	60.6	68.0	78.6	83.7	84.5	82.7	79.1	70.2	56.7	40.1		
	2%	WB	33.2	36.6	46.4	51.8	63.3	70.6	73.2	71.4	68.3	58.7	47.1	35.4		
		MCDB	34.9	39.9	52.7	61.7	73.2	79.6	82.0	79.1	75.8	65.5	53.8	37.3		
	5%	WB	30.9	33.1	41.7	48.3	59.6	67.5	70.9	69.2	65.8	55.0	42.7	33.0		
		MCDB	31.9	35.8	47.9	60.2	69.9	75.2	78.8	77.2	72.7	61.6	47.2	34.4		
	10%	WB	27.4	30.6	37.5	45.2	56.6	65.0	68.5	67.1	63.0	51.3	39.1	31.4		
		MCDB	28.2	32.7	42.0	55.8	65.5	72.6	76.7	75.1	68.9	56.3	42.8	32.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	21.3	20.8	22.0	24.3	23.3	23.8	23.5	22.1	18.0	14.9	15.9		
		MCDBR	17.1	21.8	25.1	33.1	32.9	28.3	27.0	27.7	28.3	28.7	22.3	17.8		
		MCWBR	15.6	17.5	16.8	19.0	17.2	15.0	14.0	15.1	16.7	17.9	15.6	15.5		
		MCDBR	15.7	18.6	23.3	29.7	27.2	23.3	24.1	23.3	22.2	24.6	21.5	16.2		
	5% WB	MCWBR	14.6	15.3	16.4	18.2	15.8	14.3	14.3	14.2	14.4	17.1	15.6	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.259	0.308	0.345	0.363	0.370	0.387	0.375	0.352	0.308	0.283	0.259		
	taud		2.357	2.325	2.344	2.378	2.386	2.387	2.358	2.421	2.460	2.572	2.520	2.396		
	Ebn at Noon		272	292	290	288	286	283	276	275	272	272	257	252		
	Edh at Noon		22	28	33	36	36	37	37	34	30	23	19	19		
(44) All-Sky Solar Radiation	RadAvg		452	812	1178	1473	1666	1780	1912	1617	1203	718	416	320		
	RadStd		52	76	111	149	121	134	85	122	93	98	53	33		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page